

3.2. Gaussian elimination

The three elementary row operations allow us to transform a system of equations into a different system of equations with the same solutions. We will use these elementary row operations to systematically simplify a system of equations until we can directly read off the solutions. This process is called Gaussian elimination.

Example: Solve the following linear system of equations.

$$\begin{cases} 2x_1 + 8x_2 + 4x_3 = 2 \\ 2x_1 + 5x_2 + x_3 = 5 \\ 4x_1 + 10x_2 - x_3 = 1 \end{cases}$$

To do this, we first write the corresponding augmented matrix, and we do row operations until the left side of the augmented matrix has ones in the diagonal and zeros everywhere else.

$$\left[\begin{array}{ccc|c} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array} \right] \xrightarrow{\frac{1}{2} \cdot R_1} \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array} \right] \begin{array}{l} R_2 - 2 \cdot R_1 \\ R_3 - 4 \cdot R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & -3 & -3 & 3 \\ 0 & -6 & -9 & -3 \end{array} \right] \xrightarrow{\frac{1}{-3} \cdot R_2} \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -6 & -9 & -3 \end{array} \right] \xrightarrow{R_3 + 6 \cdot R_2}$$

$$\begin{array}{ccc}
 \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -3 & -9 \end{array} \right] & \xrightarrow{\frac{1}{-3} \cdot R_3} & \left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right] \\
 & & \xrightarrow{R_1 - 4R_2} \\
 \left[\begin{array}{ccc|c} 1 & 0 & -2 & 5 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right] & \xrightarrow{\begin{array}{l} R_1 + 2R_3 \\ R_2 - R_1 \end{array}} & \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right]
 \end{array}$$

We can now translate the augmented matrix back to a system of linear equations, obtaining the following.

$$\begin{cases} x_1 = 11 \\ x_2 = -4 \\ x_3 = 3 \end{cases}$$

This is indeed a solved system of equations, with $\begin{bmatrix} 11 \\ -4 \\ 3 \end{bmatrix}$ the only solution.

The final form of the augmented matrix after doing as many simplifications as possible

is known as reduced row echelon form. In general, it will have the following shape.

$$\left[\begin{array}{cccccccccccc|c} 1 & * & * & 0 & * & 0 & 0 & * & * & * & * & * \\ \hline & & & 1 & * & 0 & 0 & * & * & * & * & * \\ & & & & & 1 & 0 & * & * & * & * & * \\ & & & & & & 1 & * & * & * & * & * \\ & & & & & & & 1 & * & * & * & * \\ & & & & & & & & & & * & * \\ & & & & & & & & & & & * \end{array} \right]$$

In the shape above there must be zeros under the horizontal line. Note that in

the example above we could have stopped after obtaining the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 4 & 2 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

because the corresponding linear system of equations is

$$\begin{cases} x_1 + 4x_2 + 2x_3 = 1 \\ x_2 + x_3 = -1 \\ x_3 = 3 \end{cases}$$

which can be solved by $x_3 = 3$ from the third equation, substituting it into the second equation to obtain $x_2 = -1 - x_3 = -1 - 3 = -4$, and substitute it into the first equation to obtain $x_1 = 1 - 4x_2 - 2x_3 = 1 + 16 - 6 = 11$. This intermediate form before fully simplifying is known as row echelon form. In general, it will have the

following shape, where again there must be zeros under the horizontal line.

$$\left[\begin{array}{cccccccccccc|c} 1 & * & * & * & * & * & * & * & * & * & * & * \\ \hline & & & 1 & * & * & * & * & * & * & * & * \\ & & & & & 1 & * & * & * & * & * & * \\ & & & & & & 1 & * & * & * & * & * \\ & & & & & & & 1 & * & * & * & * \\ & & & & & & & & & & * & * \\ & & & & & & & & & & & * \end{array} \right]$$

It is not always possible to obtain the ideal row reduced echelon form. This could

potentially indicate that we have infinitely many solutions, or no solutions at all.

Example: Solve the following linear system of equations.

$$\begin{cases} 2x_1 - 4x_2 + 6x_3 = 4 \\ 4x_1 - 2x_2 = 2 \\ -6x_1 + 6x_3 = -2 \end{cases}$$

Proceeding systematically, we will again attempt to obtain an augmented matrix

whose left side has ones in the diagonal and zeros everywhere else.

$$\left[\begin{array}{ccc|c} 2 & -4 & 6 & 4 \\ 4 & -2 & 0 & 2 \\ -6 & 0 & 6 & -2 \end{array} \right] \xrightarrow{\frac{1}{2} \cdot R_1} \left[\begin{array}{ccc|c} 1 & -2 & 3 & 2 \\ 4 & -2 & 0 & 2 \\ -6 & 0 & 6 & -2 \end{array} \right] \begin{array}{l} R_2 - 4 \cdot R_1 \\ R_3 + 6 \cdot R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 2 \\ 0 & 6 & -12 & -6 \\ 0 & -12 & 24 & 10 \end{array} \right] \xrightarrow{\frac{1}{6} \cdot R_2} \left[\begin{array}{ccc|c} 1 & -2 & 3 & 2 \\ 0 & 1 & -2 & -1 \\ 0 & -12 & 24 & 10 \end{array} \right] \xrightarrow{R_3 + 12 \cdot R_2}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 2 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & -2 \end{array} \right] \xrightarrow{-\frac{1}{2} \cdot R_3} \left[\begin{array}{ccc|c} 1 & -2 & 3 & 2 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 + 2R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2 + R_3} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

The very last step is not necessary, but it is good practice to make all

entries above the "diagonal" ones zero. Translating the augmented matrix back to a system of linear equations gives the following.

$$\begin{cases} x_1 - x_3 = 0 \\ x_2 - 2x_3 = 0 \\ 0 = 1 \end{cases}$$

Since the equation $0 = 1$ never holds true, the system has no solutions.

Example: Solve the following linear system of equations.

$$\begin{cases} -2x_1 - 2x_2 + 4x_3 = 10 \\ 5x_1 + x_2 - 2x_3 = 7 \\ x_1 - 3x_2 + 6x_3 = 27 \end{cases}$$

We again attempt to obtain an augmented matrix whose left side has ones in the diagonal and zeros everywhere else.

$$\left[\begin{array}{ccc|c} -2 & -2 & 4 & 10 \\ 5 & 1 & -2 & 7 \\ 1 & -3 & 6 & 27 \end{array} \right] \xrightarrow{\frac{1}{-2} \cdot R_1} \left[\begin{array}{ccc|c} 1 & 1 & -2 & -5 \\ 5 & 1 & -2 & 7 \\ 1 & -3 & 6 & 27 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 - 5 \cdot R_1 \\ R_3 - R_1 \end{array}}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & -5 \\ 0 & -4 & 8 & 32 \\ 0 & -4 & 8 & 32 \end{array} \right] \xrightarrow{\begin{array}{l} \frac{1}{-4} \cdot R_2 \\ \frac{1}{-4} \cdot R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 1 & -2 & -5 \\ 0 & 1 & -2 & -8 \\ 0 & 1 & -2 & -8 \end{array} \right] \xrightarrow{\begin{array}{l} R_1 - R_2 \\ R_3 - R_2 \end{array}}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

This augmented matrix translates to the following system of linear equations.

$$\begin{cases} x_1 = 3 \\ x_2 - 2x_3 = -8 \\ 0 = 0 \end{cases}$$

The third equation is irrelevant, since it provides no restrictions and no additional information at all, it is simply a mathematical truth. The first equation

gives $x_1 = 3$. The second imposes a dependency between x_2 and x_3 , which we

can write as either $x_2 = 2x_3 - 8$ or $x_3 = \frac{1}{2}x_2 + 4$. To make our life

simpler, we avoid fractions and choose $x_2 = 2x_3 - 8$. We then have the

following system of linear equations.

$$\begin{cases} x_1 = 3 \\ x_2 = 2x_3 - 8 \end{cases}$$

Since x_3 is not constrained, we set $x_3 = s$ a parameter, obtaining

$$\begin{cases} x_1 = 3 \\ x_2 = -8 + 2s \\ x_3 = s \end{cases}$$

which can also be written in the following parametric form.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ -8 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

This gives infinitely many solutions, namely all the points in the line given by the parametric form above.

In general, there are three possible reduced row echelon forms, each of them corresponding to whether the original system of equations has exactly one solution, no solutions, or infinitely many solutions.

$$\begin{bmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If the reduced row echelon form has ones in the diagonal and all other rows are zero, the original system has exactly one solution.

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

If the reduced row echelon form has a row with a one in the last column and zeros in all other columns, the original system has no solutions.

$$\begin{bmatrix} 1 & & & & \\ & 0 & & & \\ & & 1 & & \\ & & & 1 & \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

If the reduced row echelon form does not have all ones in the diagonal and does not have a row with a one in the last column and zeros in all other columns, the original system has infinitely many solutions.

To make this concept precise we use the following notion. The rank of a matrix is the number of non-zero rows in its reduced row echelon form.

Example: Find the rank of the following augmented matrices.

(i) The rank of

$$\left[\begin{array}{ccc|c} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array} \right] \text{ is 3 because } \left[\begin{array}{ccc|c} 1 & 0 & 0 & 11 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

is its reduced row echelon form, which has three non-zero rows.

(ii) The rank of

$$\left[\begin{array}{ccc|c} 2 & -4 & 6 & 4 \\ 4 & -2 & 0 & 2 \\ -6 & 0 & 6 & -2 \end{array} \right] \text{ is 3 because } \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

is its reduced row echelon form, which has three non-zero rows.

(iii) The rank of

$$\left[\begin{array}{ccc|c} -2 & -2 & 4 & 10 \\ 5 & 1 & -2 & 7 \\ 1 & -3 & 6 & 27 \end{array} \right] \text{ is 2 because } \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & -2 & -8 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

is its reduced row echelon form, which has two non-zero rows.

The existence and number of solutions can now be determined in terms of the rank of a matrix and the number of unknowns.

(a) If the rank of the augmented matrix equals the number of unknowns, the system has exactly one solution.

(b) If the rank of the augmented matrix is strictly greater than the rank of the matrix obtained by removing the last column of the augmented matrix, the system has no solutions.

(c) If the rank of the augmented matrix equals the rank of the matrix obtained by removing the last column of the augmented matrix and is strictly smaller than the number of unknowns, the system has infinitely many solutions.

Example: Determine the number of solutions of the following systems using ranks.

(i) The system

$$\begin{cases} 2x_1 + 8x_2 + 4x_3 = 2 \\ 2x_1 + 5x_2 + x_3 = 5 \\ 4x_1 + 10x_2 - x_3 = 1 \end{cases} \quad \text{with augmented matrix} \quad \left[\begin{array}{ccc|c} 2 & 8 & 4 & 2 \\ 2 & 5 & 1 & 5 \\ 4 & 10 & -1 & 1 \end{array} \right]$$

has exactly one solution because it has three unknowns and the rank of the augmented matrix is three.

(ii) The system

$$\begin{cases} 2x_1 - 4x_2 + 6x_3 = 4 \\ 4x_1 - 2x_2 = 2 \\ -6x_1 + 6x_3 = -2 \end{cases} \quad \text{with augmented matrix} \quad \left[\begin{array}{ccc|c} 2 & -4 & 6 & 4 \\ 4 & -2 & 0 & 2 \\ -6 & 0 & 6 & -2 \end{array} \right]$$

has no solutions because the rank of the augmented matrix is three and the rank of the matrix obtained by removing the last column of the augmented matrix is two:

$$\left[\begin{array}{ccc} 2 & -4 & 6 \\ 4 & -2 & 0 \\ -6 & 0 & 6 \end{array} \right] \quad \text{has reduced row echelon form} \quad \left[\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array} \right]$$

which indeed has two non-zero rows.

(iii) The system

$$\begin{cases} -2x_1 - 2x_2 + 4x_3 = 10 \\ 5x_1 + x_2 - 2x_3 = 7 \\ x_1 - 3x_2 + 6x_3 = 27 \end{cases}$$

has infinitely many solutions because the number of unknowns is three, the rank of the augmented matrix is two, and the rank of the matrix obtained by removing the last column of the augmented matrix is two;

$$\begin{bmatrix} -2 & -2 & 4 \\ 5 & 1 & -2 \\ 1 & -3 & 6 \end{bmatrix} \text{ has reduced row echelon form } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

which indeed has two non-zero rows.

→ determine linear dependence/independence